

ProGreaux® Bagasse-Based Biochar

Technical, Nutritional & Environmental Report

1. Executive Summary

ProGreaux® (PG) Bagasse-based Biochar is a 100% organic, carbon-rich soil amendment produced from 100% sugarcane bagasse in Louisiana, USA.

Based on 12 independent IBI laboratory analyses (attached), the product demonstrates consistent quality, stable carbon structure, measurable inherent nutrients (N-P-K), and extremely low heavy metal concentrations.

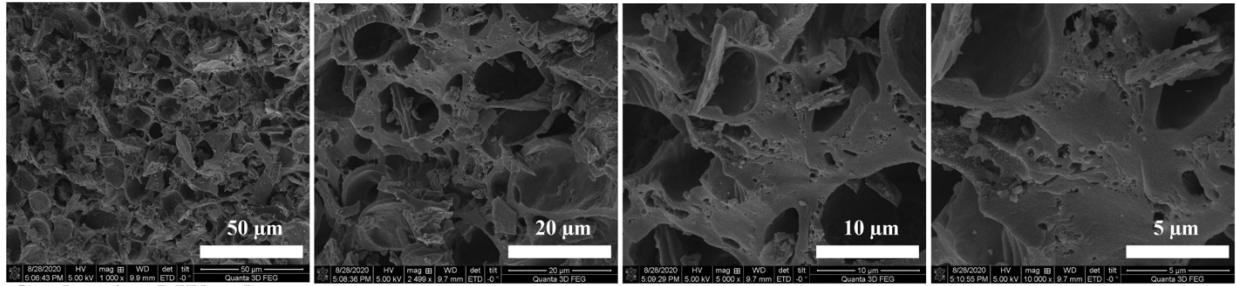
PG Bagasse-based Biochar functions simultaneously as:

- A **soil conditioner** (structure, porosity, water retention)
 - A **nutrient-retention matrix**
 - A **slow-release source of naturally occurring plant nutrients**
 - A **long-term soil carbon input**
-

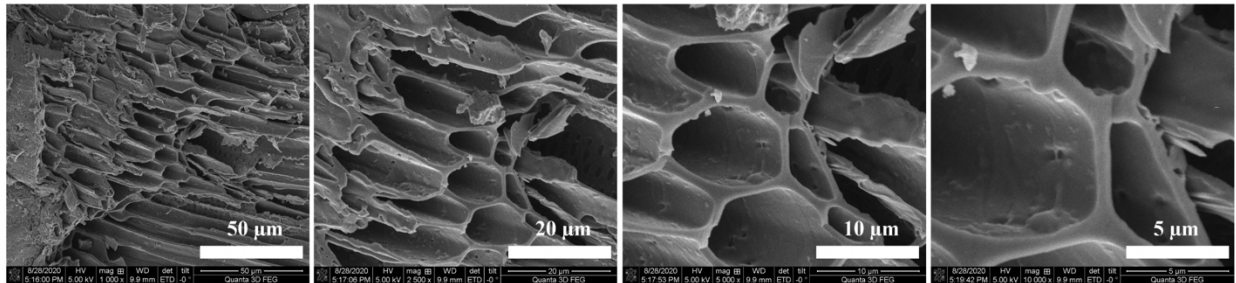
2. Feedstock & Sustainability

- Feedstock: **Untreated sugarcane bagasse**
 - Source: Agricultural by-product of cane sugar manufacturing
 - **No trees cut**
 - No treated wood, biosolids, or municipal waste
 - Renewable, annually replenished biomass
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SEM Images of carbonized wood and bagasse



Carbonized Wood



Carbonized Bagasse

Structural Differences Observed by SEM

Carbonized Wood Biochar

Observed structure

- Irregular, fractured pore network
- Predominantly **collapsed vessels and disconnected pores**
- Wide variability in pore size and shape
- More brittle carbon matrix with broken cell walls

Implications

- Lower effective pore connectivity
- Water is held primarily in **macropores**, which drain more readily
- Reduced internal surface area available for nutrient adsorption
- Minimal inherent nutrient contribution (low ash content)

Carbonized Bagasse-based Biochar

Observed structure

- Highly **ordered, honeycomb-like cellular architecture**
- Long, continuous channels aligned with original vascular fiber structure
- Smooth, intact pore walls with high interconnectivity
- Uniform pore dimensions at the 10–50 µm scale

Implications

- Significantly higher **capillary water retention**
 - Stronger nutrient adsorption and retention within pore walls
 - Enhanced microbial colonization due to protected pore habitats
 - Higher mineral ash retention (K, P, Ca, Mg) embedded in the carbon matrix
-

Functional Comparison (Soil Performance)

Property	Wood Biochar	Bagasse Biochar
Pore geometry	Random, fractured	Ordered, honeycomb
Pore connectivity	Low–moderate	High
Water-holding capacity	~2× own weight	~3–3.5× own weight
Nutrient retention	Adsorptive only	Adsorptive + inherent nutrients
Ash / mineral content	Low	Moderate–high
Microbial habitat	Limited protection	Excellent protected niches
Performance in sandy soils	Moderate	Excellent
Performance in clay soils	Moderate	Improves porosity & drainage

Why This Matters Agronomically

- **Bagasse biochar** behaves like a **biological sponge**: it stores water and nutrients during wet periods and releases them gradually during drought or nutrient stress.
- **The continuous pore channels visible in the SEM images explain why bagasse biochar:**
 - Retains more water
 - Reduces fertilizer leaching
 - Improves nutrient use efficiency

- Wood biochar, while still beneficial for carbon sequestration, lacks the same pore regularity and nutrient-bearing ash, making it primarily a structural soil conditioner rather than a combined conditioner + nutrient contributor.
-

Wood-based Biochar -v- Bagasse-based Biochar

Summary Statement

SEM imaging clearly demonstrates that bagasse-derived biochar possesses a uniquely ordered, honeycomb-like pore structure with superior pore connectivity compared to wood-derived biochar. This structural difference directly translates into higher water-holding capacity, improved nutrient retention, enhanced microbial habitat, and superior agronomic performance, particularly in sandy and degraded soils.

3. Biochar Manufacturing Process

- Oxygen-limited thermal carbonization (>500 °C)
 - No chemical activators, binders, acids, or bases
 - Syngas generated during pyrolysis is captured and oxidized for process heat
 - Solid mass reduction ~65–75%, concentrating stable carbon and mineral ash
 - All production batches meet IBI H/C molar ratio < 0.7, confirming long-term carbon stability
-

4. Physical & Chemical Characteristics

Parameter	Typical Value
Organic Carbon	~58–65% (dry basis)
Ash Content	~25–40%
H/C Molar Ratio	0.38–0.61 (≤ 0.7)
pH	~7.6–9.3
Surface Area	~215–300 m ² /g
Bulk Density	~10–23 lb/ft ³
EC	<0.35 dS/m

These properties support:

- high water-holding capacity,
 - nutrient adsorption,
 - microbial habitat.
-

5. Carbon Stability & Soil Conditioning Function

The honeycomb-like porous structure typical of bagasse-derived biochar:

- Improves soil aggregation and porosity
- Increases plant-available water
- Reduces nutrient leaching
- Provides long-term habitat for beneficial microorganisms

The carbon fraction is highly recalcitrant, supporting decades- to centuries-scale soil carbon retention.

6. Nutritional Value as a Soil Amendment

Unlike many wood-based biochar, PG Bagasse-based Biochar contains measurable, naturally occurring nutrients retained from the original biomass and mineral ash.

Average Nutrient Content (Dry Basis)

Elemental

- Nitrogen (N): **~0.64%**
 - Phosphorus (P): **~0.08%**
 - Potassium (K): **~0.55%**
-

Fertilizer Equivalent

- **N – P₂O₅ – K₂O:**
-

NPK

0.6 – 0.2 – 0.7

Secondary nutrients commonly present:

- **Calcium (Ca): ~0.8–1.5%**
 - **Magnesium (Mg): ~1.0–2.0%**
-

Important:

Nutrients are inherent, not added, and released gradually through soil biological processes. PG Biochar contains organic fertilizer, that contribute to soil fertility and nutrient efficiency.

7. Heavy Metal & Trace Element Analysis

Summary

Heavy metals in PG Bagasse-based Biochar are far below IBI, EPA Class A, and organic-use concern thresholds.

Average Heavy Metals (mg/kg, dry basis)

Element	Avg	IBI Threshold
Arsenic (As)	~0.55	13
Cadmium (Cd)	~0.31	1.4
Chromium (Cr)	~14.5	93
Cobalt (Co)	~1.5	34
Copper (Cu)	~7.4	143
Lead (Pb)	~2.7	121
Mercury (Hg)	ND	1
Nickel (Ni)	~6.1	47
Selenium (Se)	ND	2
Zinc (Zn)	~105	416

All values are one to two orders of magnitude below limits.

8. Source of Minerals & Trace Elements

Trace elements originate from:

- Naturally occurring Louisiana sugarcane field soils
 - Mineral ash retained during carbonization
 - Concentration via mass reduction (no addition)
-

9. Agronomic Performance Summary

PG Bagasse-Based Biochar:

- Improves water-holding capacity ($\approx 3\text{--}3.5\times$ its weight)
 - Reduces fertilizer runoff and leaching
 - Enhances nutrient use efficiency
 - Improves performance in sandy and heavy clay soils
 - Supports drought resilience
-

10. Regulatory & Certification Positioning

- Nonsynthetic soil amendment
 - Inherent plant nutrients
 - Heavy metals below concern thresholds
 - IBI certified
 - USDA Organic certification (pending)
 - OMRI Certified
 - Compatible with carbon removal and soil carbon accounting frameworks
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11. Intended Use

- Soil amendment and soil conditioner
 - Nutrient retention matrix
 - Long-term soil carbon input
 - Suitable for agriculture, horticulture, land reclamation, and soil remediation
-

12. Conclusion

Based on 12 independent IBI laboratory analyses, ProGreaux® Bagasse-Based biochar is a high-quality, nutrient-bearing biochar that delivers both physical soil improvement and inherent fertility benefits, while meeting the strictest industry and regulatory standards.

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Batch:
APR 23 A
CODE:
BioChar IBI

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Date Received: 4/2/2023
Sample ID: B-1-1-18-23-33
Lab ID. Number: 3030075-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	29.1		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	19.2		lb/cu ft	
Organic Carbon	59.1		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.61	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	36.6		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.67		% of total dry mass	Dry Combustion
pH value	8.43		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.155		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	4.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.5		%CaCO3	ASTM D 4373
Butane Act.	3.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	230		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
	Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic (As)	ND	13 to 100	0.48	J		< 0.5mm	85.0 percent	F
Cadmium (Cd)	0.25	1.4 to 39	0.19	J		0.5-1mm	10.5 percent	F
Chromium (Cr)	5.8	93 to 1200	0.48	J		1-2mm	3.7 percent	F
Cobalt (Co)	1.0	34 to 100	0.48	J		2-4mm	0.8 percent	F
Copper (Cu)	7.2	143 to 6000	0.48	J		4-8mm	0.0 percent	F
Lead (Pb)	2.6	121 to 300	0.19	J		8-16mm	0.0 percent	F
Molybdenum (Mo)	0.81	5 to 75	0.48	J		16-25mm	0.0 percent	F
Mercury (Hg)	ND	1 to 17	0.001	EPA 7471		25-50mm	0.0 percent	F
Nickel (Ni)	4.3	47 to 420	0.48	J		>50mm	0.0 percent	F
Selenium (Se)	ND	2 to 200	0.97	J		Basic Soil Enhancement Properties		
Zinc (Zn)	127	416 to 7400	0.97	J		Total (K)	4083 mg/kg	E
Boron (B)	5.6	Declaration	4.8	TMECC		Total (P)	686 mg/kg	E
Chlorine (Cl)	26.8	Declaration	20.0	TMECC		Ammonia (NH4-N)	9.4 mg/kg	A
Sodium (Na)	ND	Declaration	483	E		Nitrate (NO3-N)	1.7 mg/kg	A
Iron (Fe)	6769	Declaration	24.1	E		Organic (Org-N)	6696 mg/kg	Calc.
Manganese (Mn)	103	Declaration	0.48	J		Volatile Matter	13.6 percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello,

& Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Date Received: 5/22/2023
Sample ID: B-2-2-28-23-37
Lab ID. Number: 3050501-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	38.1		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.4		lb/cu ft	
Organic Carbon	61.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.37	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	34.7		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.62		% of total dry mass	Dry Combustion
pH value	7.92		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.306		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.4		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.2		%CaCO3	ASTM D 4373
Butane Act.	3.8		g/100g dry	ASTM D 5742-95
Surface Area Correlation	254		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	0.51	13 to 100	0.43	J	< 0.5mm	96.2	percent	F
Cadmium	(Cd)	0.20	1.4 to 39	0.17	J	0.5-1mm	3.0	percent	F
Chromium	(Cr)	16.4	93 to 1200	0.43	J	1-2mm	0.3	percent	F
Cobalt	(Co)	1.4	34 to 100	0.43	J	2-4mm	0.3	percent	F
Copper	(Cu)	18.9	143 to 6000	0.43	J	4-8mm	0.2	percent	F
Lead	(Pb)	3.2	121 to 300	0.17	J	8-16mm	0.0	percent	F
Molybdenum	(Mo)	5.2	5 to 75	0.43	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	5.4	47 to 420	0.43	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.85	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	78.1	416 to 7400	0.85	J	Total (K)	5418	mg/kg	E
Boron	(B)	12.8	Declaration	4.3	TMECC	Total (P)	673	mg/kg	E
Chlorine	(Cl)	121	Declaration	20.0	TMECC	Ammonia (NH4-N)	9.0	mg/kg	A
Sodium	(Na)	2764	Declaration	427	E	Nitrate (NO3-N)	2.7	mg/kg	A
Iron	(Fe)	5174	Declaration	21.4	E	Organic (Org-N)	6238	mg/kg	Calc.
Manganese	(Mn)	107	Declaration	0.43	J	Volatile Matter	10.1	percent dw	D

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Date Received: 6/8/2023
Sample ID: B-2-3-21-23-33
Lab ID. Number: 3060184-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	51.6		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	9.8		lb/cu ft	
Organic Carbon	64.3		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.50	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	25.3		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.64		% of total dry mass	Dry Combustion
pH value	8.69		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.180		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	10.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.7		%CaCO3	ASTM D 4373
Butane Act.	4.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	259		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	0.45	13 to 100	0.44	J	< 0.5mm	92.4	percent	F
Cadmium	(Cd)	0.28	1.4 to 39	0.17	J	0.5-1mm	4.8	percent	F
Chromium	(Cr)	16.0	93 to 1200	0.44	J	1-2mm	0.4	percent	F
Cobalt	(Co)	1.5	34 to 100	0.44	J	2-4mm	0.5	percent	F
Copper	(Cu)	8.2	143 to 6000	0.44	J	4-8mm	0.1	percent	F
Lead	(Pb)	2.8	121 to 300	0.17	J	8-16mm	1.7	percent	F
Molybdenum	(Mo)	1.0	5 to 75	0.44	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	7.9	47 to 420	0.44	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.87	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	57.5	416 to 7400	0.87	J	Total (K)	5338	mg/kg	E
Boron	(B)	15.1	Declaration	4.4	TMECC	Total (P)	836	mg/kg	E
Chlorine	(Cl)	184	Declaration	20.0	TMECC	Ammonia (NH4-N)	3.6	mg/kg	A
Sodium	(Na)	831	Declaration	437	E	Nitrate (NO3-N)	2.0	mg/kg	A
Iron	(Fe)	5727	Declaration	21.9	E	Organic (Org-N)	6402	mg/kg	Calc.
Manganese	(Mn)	244	Declaration	0.44	J	Volatile Matter	31.3	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello,
& Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Date Received: 8/8/2023
Sample ID: B-1-6-6-23
Lab ID. Number: 3080165-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	43.0		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	10.5		lb/cu ft	
Organic Carbon	62.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.51	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	27.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.65		% of total dry mass	Dry Combustion
pH value	7.73		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.077		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.0		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.4		%CaCO3	ASTM D 4373
Butane Act.	4.1		g/100g dry	ASTM D 5742-95
Surface Area Correlation	264		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
		Results	Max. Levels	Limit (ppm)	Method	Results	Units	Method
Arsenic	(As)	0.60	13 to 100	0.43	J	< 0.5mm	77.1 percent	F
Cadmium	(Cd)	0.31	1.4 to 39	0.17	J	0.5-1mm	18.8 percent	F
Chromium	(Cr)	18.3	93 to 1200	0.43	J	1-2mm	2.0 percent	F
Cobalt	(Co)	1.3	34 to 100	0.43	J	2-4mm	0.2 percent	F
Copper	(Cu)	6.9	143 to 6000	0.43	J	4-8mm	0.7 percent	F
Lead	(Pb)	2.6	121 to 300	0.17	J	8-16mm	1.3 percent	F
Molybdenum	(Mo)	0.64	5 to 75	0.43	J	16-25mm	0.0 percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0 percent	F
Nickel	(Ni)	8.7	47 to 420	0.43	J	>50mm	0.0 percent	F
Selenium	(Se)	ND	2 to 200	0.85	J	Basic Soil Enhancement Properties		
Zinc	(Zn)	47.5	416 to 7400	0.85	J	Total (K)	5342 mg/kg	E
Boron	(B)	ND	Declaration	4.3	TMECC	Total (P)	684 mg/kg	E
Chlorine	(Cl)	211	Declaration	20.0	TMECC	Ammonia (NH4-N)	9.7 mg/kg	A
Sodium	(Na)	ND	Declaration	427	E	Nitrate (NO3-N)	5.1 mg/kg	A
Iron	(Fe)	4594	Declaration	21.3	E	Organic (Org-N)	6478 mg/kg	Calc.
Manganese	(Mn)	98.6	Declaration	0.43	J	Volatile Matter	15.9 percent dw	D

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Date Received: 10/12/2023
Sample ID: Week 40, 2023 B-3-10-4-23
Lab ID. Number: 3100311-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	30.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	10.1		lb/cu ft	
Organic Carbon	69.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.40	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	23.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.64		% of total dry mass	Dry Combustion
pH value	8.83		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.154		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	7.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	6.5		%CaCO3	ASTM D 4373
Butane Act.	4.6		g/100g dry	ASTM D 5742-95
Surface Area Correlation	278		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
	Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic (As)	ND	13 to 100	0.50	J		< 0.5mm	57.6 percent	F
Cadmium (Cd)	0.50	1.4 to 39	0.20	J		0.5-1mm	28.4 percent	F
Chromium (Cr)	30.7	93 to 1200	0.50	J		1-2mm	13.7 percent	F
Cobalt (Co)	2.0	34 to 100	0.50	J		2-4mm	0.4 percent	F
Copper (Cu)	7.2	143 to 6000	0.50	J		4-8mm	0.0 percent	F
Lead (Pb)	2.6	121 to 300	0.20	J		8-16mm	0.0 percent	F
Molybdenum (Mo)	0.74	5 to 75	0.50	J		16-25mm	0.0 percent	F
Mercury (Hg)	ND	1 to 17	0.001	EPA 7471		25-50mm	0.0 percent	F
Nickel (Ni)	13.6	47 to 420	0.50	J		>50mm	0.0 percent	F
Selenium (Se)	ND	2 to 200	1.00	J		Basic Soil Enhancement Properties		
Zinc (Zn)	132	416 to 7400	1.00	J		Total (K)	5109 mg/kg	E
Boron (B)	42.1	Declaration	5.0	TMECC		Total (P)	923 mg/kg	E
Chlorine (Cl)	106	Declaration	20.0	TMECC		Ammonia (NH4-N)	1.0 mg/kg	A
Sodium (Na)	1114	Declaration	499	E		Nitrate (NO3-N)	1.0 mg/kg	A
Iron (Fe)	7649	Declaration	24.9	E		Organic (Org-N)	6415 mg/kg	Calc.
Manganese (Mn)	129	Declaration	0.50	J		Volatile Matter	9.1 percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Batch:
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Date Received: 10/12/2023
Sample ID: Week 39, 2023 B-4-9-27-23
Lab ID. Number: 3100312-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)		30.7	% wet wt.	ASTM D1762-84 (105c)
Bulk Density		10.1	lb/cu ft	
Organic Carbon		70.2	% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)		0.38 0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash		24.5	% of total dry mass	ASTM D-1762-84
Total Nitrogen		0.66	% of total dry mass	Dry Combustion
pH value		8.58	units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)		0.128	dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)		7.2	%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)		0.4	%CaCO3	ASTM D 4373
Butane Act.		4.3	g/100g dry	ASTM D 5742-95
Surface Area Correlation		270	m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
	Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic (As)	ND	13 to 100	0.49	J		< 0.5mm	70.2 percent	F
Cadmium (Cd)	0.53	1.4 to 39	0.20	J		0.5-1mm	26.0 percent	F
Chromium (Cr)	18.5	93 to 1200	0.49	J		1-2mm	3.4 percent	F
Cobalt (Co)	1.7	34 to 100	0.49	J		2-4mm	0.4 percent	F
Copper (Cu)	5.6	143 to 6000	0.49	J		4-8mm	0.0 percent	F
Lead (Pb)	2.9	121 to 300	0.20	J		8-16mm	0.0 percent	F
Molybdenum (Mo)	ND	5 to 75	0.49	J		16-25mm	0.0 percent	F
Mercury (Hg)	ND	1 to 17	0.001	EPA 7471		25-50mm	0.0 percent	F
Nickel (Ni)	6.7	47 to 420	0.49	J		>50mm	0.0 percent	F
Selenium (Se)	ND	2 to 200	0.99	J		Basic Soil Enhancement Properties		
Zinc (Zn)	131	416 to 7400	0.99	J		Total (K)	5154 mg/kg	E
Boron (B)	59.4	Declaration	4.9	TMECC		Total (P)	902 mg/kg	E
Chlorine (Cl)	69	Declaration	20.0	TMECC		Ammonia (NH4-N)	1.6 mg/kg	A
Sodium (Na)	564	Declaration	493	E		Nitrate (NO3-N)	2.0 mg/kg	A
Iron (Fe)	6375	Declaration	24.7	E		Organic (Org-N)	6571 mg/kg	Calc.
Manganese (Mn)	116	Declaration	0.49	J		Volatile Matter	13.3 percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello,
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J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Date Received: 11/22/2023
Sample ID: B-11-14-23-2
Lab ID. Number: 3110472-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	32.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.9		lb/cu ft	
Organic Carbon	62.0		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.58	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	33.9		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.72		% of total dry mass	Dry Combustion
pH value	9.18		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.319		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.9		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	1.6		%CaCO3	ASTM D 4373
Butane Act.	5.2		g/100g dry	ASTM D 5742-95
Surface Area Correlation	298		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	0.36	13 to 100	0.33	J	< 0.5mm	84.9	percent	F
Cadmium	(Cd)	0.24	1.4 to 39	0.13	J	0.5-1mm	12.6	percent	F
Chromium	(Cr)	7.5	93 to 1200	0.33	J	1-2mm	2.1	percent	F
Cobalt	(Co)	1.2	34 to 100	0.33	J	2-4mm	0.3	percent	F
Copper	(Cu)	8.8	143 to 6000	0.33	J	4-8mm	0.0	percent	F
Lead	(Pb)	2.6	121 to 300	0.13	J	8-16mm	0.0	percent	F
Molybdenum	(Mo)	0.58	5 to 75	0.33	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	3.4	47 to 420	0.33	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.65	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	54.7	416 to 7400	0.65	J	Total (K)	8438	mg/kg	E
Boron	(B)	10.7	Declaration	3.3	TMECC	Total (P)	899	mg/kg	E
Chlorine	(Cl)	152	Declaration	20.0	TMECC	Ammonia (NH4-N)	10.2	mg/kg	A
Sodium	(Na)	342	Declaration	327	E	Nitrate (NO3-N)	2.8	mg/kg	A
Iron	(Fe)	5163	Declaration	16.3	E	Organic (Org-N)	7182	mg/kg	Calc.
Manganese	(Mn)	96.6	Declaration	0.33	J	Volatile Matter	10.0	percent dw	D

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Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Date Received: 12/27/2024
Sample ID: B-12-7-23-1
Lab ID. Number: 3120549-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	24.4		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	22.9		lb/cu ft	
Organic Carbon	52.1		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.44	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	40.3		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.46		% of total dry mass	Dry Combustion
pH value	9.27		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.221		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	11.4		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.7		%CaCO3	ASTM D 4373
Butane Act.	5.0		g/100g dry	ASTM D 5742-95
Surface Area Correlation	292		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	0.73	13 to 100	0.36	J	< 0.5mm	90.5	percent	F
Cadmium	(Cd)	0.30	1.4 to 39	0.14	J	0.5-1mm	6.4	percent	F
Chromium	(Cr)	12.1	93 to 1200	0.36	J	1-2mm	2.5	percent	F
Cobalt	(Co)	1.7	34 to 100	0.36	J	2-4mm	0.5	percent	F
Copper	(Cu)	9.5	143 to 6000	0.36	J	4-8mm	0.0	percent	F
Lead	(Pb)	3.4	121 to 300	0.14	J	8-16mm	0.0	percent	F
Molybdenum	(Mo)	0.76	5 to 75	0.36	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	5.7	47 to 420	0.36	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.71	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	75.7	416 to 7400	0.71	J	Total (K)	6643	mg/kg	E
Boron	(B)	9.3	Declaration	3.6	TMECC	Total (P)	990	mg/kg	E
Chlorine	(Cl)	100	Declaration	20.0	TMECC	Ammonia (NH4-N)	3.9	mg/kg	A
Sodium	(Na)	ND	Declaration	357	E	Nitrate (NO3-N)	1.2	mg/kg	A
Iron	(Fe)	6669	Declaration	17.9	E	Organic (Org-N)	4603	mg/kg	Calc.
Manganese	(Mn)	130	Declaration	0.36	J	Volatile Matter	17.2	percent dw	D

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Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello,
& Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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CODE:
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Date Received: 3/21/2024
Sample ID: B-1-3-6-24
Lab ID. Number: 4030435-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	28.3		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	14.6		lb/cu ft	
Organic Carbon	63.5		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.55	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	29.5		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.69		% of total dry mass	Dry Combustion
pH value	8.52		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.180		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.2		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	1.9		%CaCO3	ASTM D 4373
Butane Act.	2.9		g/100g dry	ASTM D 5742-95
Surface Area Correlation	225		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
	Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic (As)	0.59	13 to 100	0.48	J		< 0.5mm	77.8 percent	F
Cadmium (Cd)	0.33	1.4 to 39	0.19	J		0.5-1mm	17.7 percent	F
Chromium (Cr)	7.9	93 to 1200	0.48	J		1-2mm	4.2 percent	F
Cobalt (Co)	1.5	34 to 100	0.48	J		2-4mm	0.4 percent	F
Copper (Cu)	6.2	143 to 6000	0.48	J		4-8mm	0.0 percent	F
Lead (Pb)	2.7	121 to 300	0.19	J		8-16mm	0.0 percent	F
Molybdenum (Mo)	0.7	5 to 75	0.48	J		16-25mm	0.0 percent	F
Mercury (Hg)	ND	1 to 17	0.001	EPA 7471		25-50mm	0.0 percent	F
Nickel (Ni)	4.2	47 to 420	0.48	J		>50mm	0.0 percent	F
Selenium (Se)	ND	2 to 200	0.96	J		Basic Soil Enhancement Properties		
Zinc (Zn)	170	416 to 7400	0.96	J		Total (K)	5330 mg/kg	E
Boron (B)	6.4	Declaration	4.8	TMECC		Total (P)	869 mg/kg	E
Chlorine (Cl)	116	Declaration	20.0	TMECC		Ammonia (NH4-N)	99.0 mg/kg	A
Sodium (Na)	ND	Declaration	482	E		Nitrate (NO3-N)	2.8 mg/kg	A
Iron (Fe)	7787	Declaration	24.1	E		Organic (Org-N)	6777 mg/kg	Calc.
Manganese (Mn)	99.0	Declaration	0.48	J		Volatile Matter	16.9 percent dw	D

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Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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CODE:
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Date Received: 4/4/2024
Sample ID: B-1-2-8-24
Lab ID. Number: 4040126-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated: Range	Units	Method
Moisture (time of analysis)	27.6	% wet wt.	ASTM D1762-84 (105c)
Bulk Density	13.8	lb/cu ft	
Organic Carbon	62.0	% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.52 0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	32.7	% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.71	% of total dry mass	Dry Combustion
pH value	8.30	units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.174	dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	8.1	%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.2	%CaCO3	ASTM D 4373
Butane Act.	2.6	g/100g dry	ASTM D 5742-95
Surface Area Correlation	217	m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	ND	13 to 100	0.39	J	< 0.5mm	75.4	percent	F
Cadmium	(Cd)	0.19	1.4 to 39	0.16	J	0.5-1mm	18.9	percent	F
Chromium	(Cr)	4.3	93 to 1200	0.39	J	1-2mm	5.3	percent	F
Cobalt	(Co)	0.84	34 to 100	0.39	J	2-4mm	0.4	percent	F
Copper	(Cu)	4.1	143 to 6000	0.39	J	4-8mm	0.0	percent	F
Lead	(Pb)	1.6	121 to 300	0.16	J	8-16mm	0.0	percent	F
Molybdenum	(Mo)	ND	5 to 75	0.39	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	2.4	47 to 420	0.39	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.78	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	94.1	416 to 7400	0.78	J	Total (K)	2974	mg/kg	E
Boron	(B)	4.3	Declaration	3.9	TMECC	Total (P)	431	mg/kg	E
Chlorine	(Cl)	434	Declaration	20.0	TMECC	Ammonia (NH4-N)	46.7	mg/kg	A
Sodium	(Na)	ND	Declaration	388	E	Nitrate (NO3-N)	2.8	mg/kg	A
Iron	(Fe)	4109	Declaration	19.4	E	Organic (Org-N)	7099	mg/kg	Calc.
Manganese	(Mn)	54.2	Declaration	0.39	J	Volatile Matter	23.4	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
J EPA3050B/EPA 6020

Analyst: Nik Zumberge



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Date Received: 4/4/2024
Sample ID: B-1-1-19-24
Lab ID. Number: 4040127-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)		27.1	% wet wt.	ASTM D1762-84 (105c)
Bulk Density		14.1	lb/cu ft	
Organic Carbon		59.6	% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)		0.55 0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash		28.9	% of total dry mass	ASTM D-1762-84
Total Nitrogen		0.70	% of total dry mass	Dry Combustion
pH value		8.51	units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)		0.152	dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)		13.8	%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)		0.0	%CaCO3	ASTM D 4373
Butane Act.		2.5	g/100g dry	ASTM D 5742-95
Surface Area Correlation		213	m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution		
		Results	Max. Levels	Limit (ppm)	Method	Results	Units	Method
Arsenic	(As)	0.53	13 to 100	0.42	J	< 0.5mm	70.5 percent	F
Cadmium	(Cd)	0.30	1.4 to 39	0.17	J	0.5-1mm	20.3 percent	F
Chromium	(Cr)	7.5	93 to 1200	0.42	J	1-2mm	8.6 percent	F
Cobalt	(Co)	1.42	34 to 100	0.42	J	2-4mm	0.5 percent	F
Copper	(Cu)	6.3	143 to 6000	0.42	J	4-8mm	0.0 percent	F
Lead	(Pb)	2.7	121 to 300	0.17	J	8-16mm	0.0 percent	F
Molybdenum	(Mo)	0.69	5 to 75	0.42	J	16-25mm	0.0 percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0 percent	F
Nickel	(Ni)	4.3	47 to 420	0.42	J	>50mm	0.0 percent	F
Selenium	(Se)	ND	2 to 200	0.84	J	Basic Soil Enhancement Properties		
Zinc	(Zn)	132	416 to 7400	0.84	J	Total (K)	5272 mg/kg	E
Boron	(B)	6.5	Declaration	4.2	TMECC	Total (P)	806 mg/kg	E
Chlorine	(Cl)	100	Declaration	20.0	TMECC	Ammonia (NH4-N)	8.0 mg/kg	A
Sodium	(Na)	ND	Declaration	420	E	Nitrate (NO3-N)	1.7 mg/kg	A
Iron	(Fe)	6973	Declaration	21.0	E	Organic (Org-N)	6956 mg/kg	Calc.
Manganese	(Mn)	98.3	Declaration	0.42	J	Volatile Matter	22.5 percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson
D ASTM D1762-84
E EPA3050B/EPA 6010
F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello, & Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area
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Analyst: Nik Zumberge



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CODE:
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Date Received: 4/25/2024
Sample ID: B-4-4-15-24
Lab ID. Number: 4040648-01

International BioChar Initiative (IBI) Laboratory Tests for Certification Program

	Dry Basis Unless Stated:	Range	Units	Method
Moisture (time of analysis)	39.9		% wet wt.	ASTM D1762-84 (105c)
Bulk Density	16.3		lb/cu ft	
Organic Carbon	59.9		% of total dry mass	Dry Combust-ASTM D 4373
Hydrogen/Carbon (H:C)	0.46	0.7 Max	Molar Ratio	H dry combustion/C(above)
Total Ash	32.8		% of total dry mass	ASTM D-1762-84
Total Nitrogen	0.61		% of total dry mass	Dry Combustion
pH value	7.59		units	4.11USCC:dil. Rajkovich
Electrical Conductivity (EC20 w/w)	0.141		dS/m	4.10USCC:dil. Rajkovich
Liming (neut. Value as-CaCO3)	6.7		%CaCO3	AOAC 955.01
Carbonates (as-CaCO3)	0.6		%CaCO3	ASTM D 4373
Butane Act.	3.2		g/100g dry	ASTM D 5742-95
Surface Area Correlation	235		m2/g dry	G

All units mg/kg dry unless stated:		Range of		Reporting		Particle Size Distribution			
		Results	Max. Levels	Limit (ppm)	Method		Results	Units	Method
Arsenic	(As)	0.56	13 to 100	0.43	J	< 0.5mm	72.9	percent	F
Cadmium	(Cd)	0.36	1.4 to 39	0.17	J	0.5-1mm	16.3	percent	F
Chromium	(Cr)	10.8	93 to 1200	0.43	J	1-2mm	8.8	percent	F
Cobalt	(Co)	1.6	34 to 100	0.43	J	2-4mm	1.4	percent	F
Copper	(Cu)	5.8	143 to 6000	0.43	J	4-8mm	0.6	percent	F
Lead	(Pb)	2.6	121 to 300	0.17	J	8-16mm	0.0	percent	F
Molybdenum	(Mo)	0.87	5 to 75	0.43	J	16-25mm	0.0	percent	F
Mercury	(Hg)	ND	1 to 17	0.001	EPA 7471	25-50mm	0.0	percent	F
Nickel	(Ni)	6.0	47 to 420	0.43	J	>50mm	0.0	percent	F
Selenium	(Se)	ND	2 to 200	0.85	J	Basic Soil Enhancement Properties			
Zinc	(Zn)	134	416 to 7400	0.85	J	Total (K)	5226	mg/kg	E
Boron	(B)	8.6	Declaration	4.3	TMECC	Total (P)	862	mg/kg	E
Chlorine	(Cl)	58.6	Declaration	20.0	TMECC	Ammonia (NH4-N)	2.5	mg/kg	A
Sodium	(Na)	512	Declaration	427	E	Nitrate (NO3-N)	1.3	mg/kg	A
Iron	(Fe)	7866	Declaration	21.4	E	Organic (Org-N)	6097	mg/kg	Calc.
Manganese	(Mn)	110	Declaration	0.43	J	Volatile Matter	15.2	percent dw	D

* "ND" stands for "not detected" which means the result is below the reporting limit.

Method A Rayment & Higginson

D ASTM D1762-84

E EPA3050B/EPA 6010

F ASTM D 2862 Granular

G Butane Activity Surface Area Correlation Based on McLaughlin, Shields, Jagiello,
& Thiele's 2012 paper: Analytical Options for Biochar Adsorption and Surface Area

J EPA3050B/EPA 6020

Analyst: Nik Zumberge

